CHAPTER 2

Limits of sequences

S14, S16, S19, S20

Objectives

Distinguish convergence from divergence; calculate a limit; use comparison, the squeeze theorem and monotone convergence.

Course notes

Definitions

$u_n \to \ell \in \mathbb{R}$ means that every open interval containing ℓ contains all the terms from some index onwards. We say that (u_n) converges to ℓ. $u_n \to +\infty$ means that for every real number A, there exists an index N such that $n \geq N \Rightarrow u_n \geq A$. $u_n \to -\infty$ means that for every real number A, there exists N such that $n \geq N \Rightarrow u_n \leq A$.

A sequence may have no limit, even an infinite one: $(-1)^n$ oscillates between -1 and 1. Every convergent sequence is bounded, but the converse is false.

Operations and comparisons

For two convergent sequences with limits a, b, the sum tends to $a+b$, the product to ab and the quotient to $\frac{a}{b}$ if $b \neq 0$. The rules involving infinity are used by keeping track of the signs. The forms $+\infty - \infty$, $0 \times \infty$, $\frac{\infty}{\infty}$, $\frac{0}{0}$ do not allow a direct conclusion.

Example. $\frac{3n^2-n+1}{2n^2+4} = \frac{3-\frac{1}{n}+\frac{1}{n^2}}{2+\frac{4}{n^2}} \to \frac{3}{2}$. For $\sqrt{n^2+n}-n$, multiplying by the conjugate gives $\frac{1}{\sqrt{1+\frac{1}{n}}+1} \to \frac{1}{2}$.

Theorems

If $u_n \leq v_n$ from some index onwards and $u_n \to +\infty$, then $v_n \to +\infty$. If $a_n \leq u_n \leq b_n$ from some index onwards and $a_n, b_n \to \ell$, then $u_n \to \ell$ (squeeze theorem). Every increasing sequence that is bounded above converges. Every decreasing sequence that is bounded below converges. This theorem is accepted without proof.

Proof by comparison. For every A, the terms of u exceed A beyond an index N_1. The inequality $v_n \geq u_n$ holds beyond an index N_2. For $n \geq \max(N_1, N_2)$, $v_n \geq A$.

Proof: increasing and not bounded above. For every A, the lack of an upper bound provides N such that $u_N > A$. By monotonicity, $n \geq N \Rightarrow u_n \geq u_N > A$, hence $u_n \to +\infty$. Being unbounded alone is not enough.

Geometric sequences

$$q^n \longrightarrow \begin{cases} 0 & \text{if } -1 < q < 1, \\ 1 & \text{if } q = 1, \\ +\infty & \text{if } q > 1. \end{cases}$$

If $q \leq -1$, (q^n) has no limit: alternation for $q = -1$; for $q < -1$, the even and odd subsequences have opposite signs and unbounded moduli.

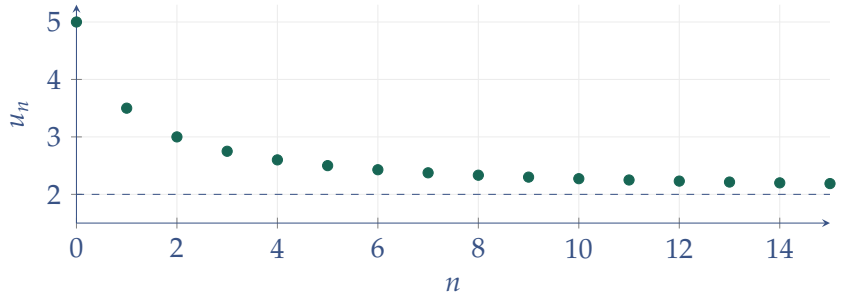
Bernoulli and the geometric limit. Fix $a \geq 0$. Let us prove $P(n)$: "$(1+a)^n \geq 1 + na$". At index 0, both sides equal 1. Fix $n \geq 0$ and assume $P(n)$ is true. Since $1 + a > 0$,

$$(1+a)^{n+1} \geq (1+na)(1+a) = 1 + (n+1)a + na^2 \geq 1 + (n+1)a.$$

This is $P(n+1)$ since $na^2 \geq 0$. By induction, the inequality is true for every $n \geq 0$. If $q > 1$, take $a = q - 1 > 0$ and compare with $1 + na$. If $0 < q < 1$, apply this result to $\frac{1}{q}$, then take reciprocals. For $-1 < q < 0$, $|q^n| = |q|^n \to 0$ and the squeeze theorem concludes. The case $q = 0$ is handled from index 1 onwards.

Recurrence sequence

For $u_{n+1} = f(u_n)$, establish a bound and monotonicity to prove convergence. If f is continuous at the limit ℓ, passing to the limit gives $\ell = f(\ell)$ (continuity is studied in chapter 4). Solving this equation never proves, on its own, that the sequence converges.



The points get closer to 2 without reaching it: $u_n = 2 + \frac{3}{n+1}$.

Research activity

A limit is not a last term — 30 min.

Consider the sequence (u_n) defined on $\mathbb{N}$ by $u_n = 2 + \frac{3}{n+1}$.

1. Calculate u_0, u_9 and u_{99}. Is there an integer n such that $u_n = 2$? Justify.
2. Determine the smallest integer N such that, for every integer $n \geq N$, $|u_n - 2| < 0.01$.
3. Let $\varepsilon > 0$. Give, in terms of ε, an integer N such that $|u_n - 2| < \varepsilon$ for every $n \geq N$.
4. Define on $\mathbb{N}$ the sequence (v_n) by $v_n = 2 + \frac{(-1)^n}{n+1}$. Prove that it converges to 2 and that it is not monotonic. What does this example show?

Progressive exercises

E1 · Practice. Define on $\mathbb{N}$ the sequences (u_n) and (v_n) by $u_n = \frac{4n-1}{n+2}$ and $v_n = n^2 - 7n$. Determine their limits, justifying the transformations used.

E2 · Practice. Define $u_n = \sqrt{n+1} - \sqrt{n}$ for every $n \in \mathbb{N}$. Using the conjugate, determine the limit of (u_n).

E3 · Practice. Define $u_n = \frac{\cos n}{n}$ for every integer $n \geq 1$. Bound u_n above and below, then deduce the limit of (u_n).

E4 · Practice. The sequence (u_n) is increasing and satisfies $u_n < 5$ for every $n \in \mathbb{N}$. Justify that it converges and state what can be asserted about its limit. Is it necessarily equal to 5? Justify.

E5 · Practice. Define $u_n = 3 - 2(-0.6)^n$ for every $n \in \mathbb{N}$. Determine the limit of (u_n). Is this sequence monotonic? Justify.

Textbook or past-paper work

Barbazo, p. 62, exercises 17 and 24. In S16: exercise 17, to resolve an indeterminate form by dividing by a power of n. At home after S20: exercise 24, to write up an application of the squeeze theorem.

Homework

Homework 2 — Approximating a root — 50 min.

Consider $u_0 = 0$ and, for every $n \in \mathbb{N}$, $u_{n+1} = \sqrt{2 + u_n}$. Questions 1, 2 and 4 are to be done after S20; question 3 will be revisited after the lesson on continuity in S31.

1. Prove by induction that all the terms are defined and that $0 \leq u_n \leq 2$ for every $n \in \mathbb{N}$.
2. Prove by induction that $u_{n+1} \geq u_n$ for every $n \in \mathbb{N}$. Deduce that (u_n) converges.
3. Using the continuity of the function $x \mapsto \sqrt{2+x}$, determine the limit of (u_n).
4. Prove that $0 \leq 2 - u_{n+1} \leq \frac{2-u_n}{2}$ for every $n \in \mathbb{N}$. Deduce an upper bound for $2 - u_n$ in terms of n, then an integer N such that $0 \leq 2 - u_n \leq 10^{-3}$ for every $n \geq N$.

Teaching notes and solutions

Teacher notes

Test 2 comes before S19 and S20: assess in it the definitions and the operations already met, without requiring monotone convergence, Bernoulli or the squeeze theorem. Draw horizontal bands around the limit and have students look for an index from which all the terms stay inside them. Require convergence before solving the fixed-point equation. Show $(-1)^n$ to separate "bounded" from "convergent".

Solutions to the activity

1. $u_0 = 5$, $u_9 = 2.3$ and $u_{99} = 2.03$. Since $\frac{3}{n+1} > 0$ at every index, $u_n > 2$: no term equals 2.
2. $|u_n - 2| < 0.01 \iff \frac{3}{n+1} < 0.01 \iff n+1 > 300 \iff n \geq 300$. The first suitable index is therefore $N = 300$. At index 299, the gap is exactly 0.01 and the strict inequality is not satisfied.
3. For $\varepsilon > 0$, take $N = \lceil \frac{3}{\varepsilon} \rceil$, the smallest integer greater than or equal to $\frac{3}{\varepsilon}$. If $n \geq N$, then $n+1 > \frac{3}{\varepsilon}$, hence $|u_n - 2| < \varepsilon$. This choice works, even when it is not minimal.
4. $|v_n - 2| = \frac{1}{n+1} \to 0$, hence $v_n \to 2$. But $v_0 = 3$, $v_1 = \frac{3}{2}$ and $v_2 = \frac{7}{3}$: we observe a decrease then an increase. This convergent sequence is not monotonic.

E1. For $n \geq 1$, $\frac{4n-1}{n+2} = \frac{4-\frac{1}{n}}{1+\frac{2}{n}} \to 4$, since the denominator tends to $1 \neq 0$. Similarly $n^2 - 7n = n^2(1 - \frac{7}{n})$; the first factor tends to $+\infty$ and the second to $1 > 0$, so the product tends to $+\infty$.

E2. The conjugate gives $\sqrt{n+1} - \sqrt{n} = \frac{(n+1)-n}{\sqrt{n+1}+\sqrt{n}} = \frac{1}{\sqrt{n+1}+\sqrt{n}}$. The denominator tends to $+\infty$, so the difference tends to 0.

E3. For $n \geq 1$, $-1 \leq \cos n \leq 1$ and $n > 0$, so $-\frac{1}{n} \leq u_n \leq \frac{1}{n}$. Both bounds tend to 0; the squeeze theorem gives $u_n \to 0$.

E4. The sequence is increasing and bounded above by 5, so it converges. By passing to the limit in $u_n \leq 5$, its limit satisfies $\ell \leq 5$. It is not necessarily equal to 5: $u_n = 2 - \frac{1}{n+1}$ is increasing, bounded above by 5 and converges to 2.

E5. Since $|-0.6| < 1$, $(-0.6)^n \to 0$, so $u_n \to 3$. However $u_0 = 1$, $u_1 = 4.2$ and $u_2 = 2.28$: the sequence rises then falls and is not monotonic.

Solutions to Homework 2

1. Let $P(n)$ be the statement "u_n is defined and $0 \leq u_n \leq 2$". It is true at index 0 since $u_0 = 0$. Fix $n \geq 0$ and assume $P(n)$ is true. We have $2 \leq 2 + u_n \leq 4$, so $u_{n+1} = \sqrt{2+u_n}$ is defined and $\sqrt{2} \leq u_{n+1} \leq 2$. Thus $P(n+1)$ is true. By induction, all the terms belong to $[0,2]$.
2. Let $R(n)$ be the statement "$u_{n+1} \geq u_n$". For $n = 0$, $u_1 = \sqrt{2} \geq 0 = u_0$. Fix $n \geq 0$ and assume $R(n)$ is true. The fact that $x \mapsto \sqrt{2+x}$ is increasing on $[0,2]$ gives

$$u_{n+2} = \sqrt{2+u_{n+1}} \geq \sqrt{2+u_n} = u_{n+1}.$$

   So $R(n+1)$ is true. By induction, (u_n) is increasing. It is bounded above by 2, so it converges to a real number $\ell \in [0,2]$.
3. The function $x \mapsto \sqrt{2+x}$ is continuous at ℓ. By passing to the limit, $\ell = \sqrt{2+\ell}$. Since $\ell \geq 0$, this is equivalent to $\ell^2 - \ell - 2 = 0$, that is $(\ell - 2)(\ell + 1) = 0$. The root -1 is excluded; hence $\ell = 2$.

4. Using the conjugate and $u_n \leq 2$,

$$0 \leq 2 - u_{n+1} = \frac{2 - u_n}{2 + \sqrt{2 + u_n}} \leq \frac{2 - u_n}{2}.$$

Let $T(n)$ be the statement "$0 \leq 2 - u_n \leq \frac{2}{2^n}$". At index 0, $2 - u_0 = 2$. If $T(n)$ is true for a fixed integer $n \geq 0$, then

$$0 \leq 2 - u_{n+1} \leq \frac{1}{2}\frac{2}{2^n} = \frac{2}{2^{n+1}}.$$

The base case and the inductive step prove $T(n)$ at every index. Since $\frac{2}{2^{11}} = \frac{1}{1024} < 10^{-3}$, $n = 11$ suffices. This is an index guaranteed by an upper bound, with no claim of minimality. This question can be done before question 3: the number 2 already appears in the bounds.

Solutions to the textbook or past-paper work

Exercise 17. The quotients are at first indeterminate forms. For $n \geq 1$,

$$u_n = \frac{2n - 4}{n^2 + 1} = \frac{2\left(1 - \frac{2}{n}\right)}{n\left(1 + \frac{1}{n^2}\right)} \longrightarrow 0.$$

Similarly,

$$v_n = \frac{2 - 5n}{4n + 7} = \frac{\frac{2}{n} - 5}{4 + \frac{7}{n}} \longrightarrow -\frac{5}{4}, \qquad w_n = n\frac{-1 - \frac{10}{n^2} + \frac{4}{n^3}}{2 + \frac{3}{n} + \frac{1}{n^2}} \longrightarrow -\infty.$$

The last quotient tends to $-\frac{1}{2}$, and the factor n to $+\infty$.

Exercise 24. For every $n \geq 1$, $-1 \leq (-1)^n \leq 1$. Since $n^2 > 0$, we obtain $-\frac{1}{n^2} \leq u_n \leq \frac{1}{n^2}$. Both bounds tend to 0; the squeeze theorem gives $u_n \to 0$.